

Portfolio Optimization & Asset Allocation Tool

Excel | Power Query | VBA | Solver | Financial Modeling

Automated Portfolio Construction & Analysis Tool

PROJECT DOCUMENTATION

Demo version | Last Updated: 30/09/2026 | by Ilona Gavaille

MODERN PORTFOLIO THEORY

Markowitz Framework

Risk · Return · Diversification · Portfolio Optimization · Efficient Frontier

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1. Project Overview

This project is an **automated Excel-based tool for portfolio construction and analysis**. It allows users to compare portfolio allocations, evaluate risk and expected return, and identify optimized portfolios under defined constraints. The model is based on **Markowitz's Modern Portfolio Theory** and combines market data, financial theory and quantitative analysis, with the workflow automated through Power Query, VBA and Excel Solver.

The current version is a **demo version** that provides a functional framework and can be further developed into a more comprehensive tool for personal or professional investment analysis.

For instructions on using the workbook, please refer to the [How to Use](#) guide included with the project.

2. Main Features

Market Data

- Historical daily stock prices retrieved through the Financial Modeling Prep API
- Automated data import and transformation using Power Query
- VBA-based data refresh

Portfolio Analysis

- Daily returns
- Annualized return and annualized volatility
- Correlation and covariance matrix
- Sharpe ratio

Portfolio Optimization

- Minimum and maximum asset weights
- 100% total allocation constraint
- 12 portfolios generated using Excel Solver
- Efficient frontier
- Capital Allocation Line
- Optimal portfolio
- Minimum Variance portfolio

Dashboard

- Dynamic portfolio selection
- Asset allocation
- Risk and return indicators
- Cumulative performance
- Drawdown analysis
- Comparison with S&P 500 benchmark
- Efficient frontier
- Live market data

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3. Portfolio Universe

The current model uses eight predefined stocks:

Ticker	Company	Sector
AAPL	Apple	Computers, Phones & Household Electronics
MSFT	Microsoft	Software & IT Services
JPM	JPMorgan Chase	Financial Services
XOM	Exxon Mobil	Oil & Gas
JNJ	Johnson & Johnson	Pharmaceuticals
AMZN	Amazon	Diversified Retail
KO	Coca-Cola	Beverages
WMT	Walmart	Food & Drug Retailing

Due to the limitations of the API's free plan, the **assets are predefined** rather than selected dynamically by the user.

Since the current investment universe consists exclusively of equities, the selection aims to provide **sector diversification within the portfolio**.

The S&P500 is used separately as a benchmark and is not included in the optimization universe.

4. Methodology

4.1 Market Data

Historical daily closing prices are retrieved through the **Financial Modeling Prep API** and processed using Power Query. The data is then used to calculate daily returns and the statistical inputs required for portfolio optimization.

The **API freemium plan** used for this version has **limitations** regarding:

- historical data coverage;
- available financial instruments;
- the length of the available historical period.

The current setup provides access to american stocks and companies but does not provide the ETFs or bonds required for a broader multi-asset optimization model.

4.2 Daily Returns

Daily returns are calculated using:

$$R_t = (P_t - P_{t-1}) / P_{t-1}$$

where:

- P_t = closing price at time t
- P_{t-1} = previous closing price

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4.3 Expected Return

The model uses **historical average daily returns as a proxy for expected future returns**.

The annualized expected return is calculated using simple arithmetic annualization:

$$\text{Expected Annual Return} = \text{Average}(\text{Daily Returns}) \times 252$$

4.4 Correlation and Covariance

The model calculates the correlation and covariance between the selected assets.

These measures are used to understand how assets move relative to each other and to determine the overall portfolio risk.

The covariance matrix is then used directly in the portfolio volatility calculation and optimization process.

4.5 Portfolio Risk

Portfolio volatility is calculated using the covariance matrix:

$$\sigma_p = \sqrt{(w^T \Sigma w) \times \sqrt{252}}$$

where:

- w = vector of portfolio weights
- Σ = covariance matrix of asset returns

This captures the interaction between assets and allows the model to account for diversification effects.

5. Portfolio Optimization

The optimization is performed using **Excel Solver**.

The model searches for portfolio weights that satisfy the defined constraints while optimizing a specific objective.

Portfolio Constraints

- Total portfolio weight = **100%**
- Asset weights $\geq$ **0%**
- User-defined minimum and maximum weights
- No short selling
- 100% of capital allocated to risky assets

5.1 Efficient Frontier

The model generates **12 portfolios with different target-return levels**.

For each target return, **Solver minimizes portfolio volatility** while respecting the allocation constraints.

The resulting portfolios form the model's efficient frontier.

This allows the user to observe the trade-off between expected return and portfolio risk.

5.2 Optimal Portfolio

The Optimal Portfolio is the portfolio that **maximizes the Sharpe ratio** under the model's constraints. It represents the portfolio with the highest risk-adjusted return within the constraints defined by the user.

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma_p$$

- R_p = portfolio expected return
- R_f = risk-free rate
- σ_p = portfolio volatility

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6. Risk-Free Rate

The risk-free rate is **manually entered by the user** according to the purpose and geographical scope of the analysis. Possible references include:

- 10-Year U.S. Treasury yield
- U.S. Treasury Bill rate
- another relevant government or risk-free benchmark

The choice of proxy should be consistent with the investment horizon and market being analyzed.

7. Dashboard

The Dashboard provides the user-facing interface for exploring the portfolios generated by the model.

Once a portfolio is selected, the Dashboard dynamically displays:

- Asset allocation
- Expected return
- Volatility
- Sharpe ratio
- Maximum drawdown
- Efficient frontier
- Capital Allocation Line
- Performance relative to SPY
- Live market data

8. Main Assumptions

Assumption	Current Implementation
Trading days	252 days per year
Return annualization	Average daily return × 252
Portfolio allocation	100% invested
Short selling	Not allowed
Asset weights	Non-negative and subject to minimum/maximum constraints
Risk-free rate	Manually entered
Expected return	Based on historical returns
Asset universe	8 predefined stocks
Transaction costs	Not included
Taxes	Not included
Market impact	Not included

These assumptions keep the model focused on the core portfolio construction framework.

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9 . Limitations

Data Limitations

The API's free plan limits historical coverage and the available investment universe. The optimization is therefore currently **restricted to american stocks** and does not include ETFs, bonds or other asset classes. Therefore, the eight assets are **selected in advance**. The current user interface does not allow users to create a completely custom investment universe.

Expected Return Estimation

Historical returns are used as a **proxy for future expected returns**. This is a significant modelling assumption because future market conditions may differ from the historical period used in the analysis.

Risk-Free Allocation

The current Optimal Portfolio assumes that all capital is invested in risky assets. The allocation between risky assets and the risk-free asset is not yet optimized.

Simplified Portfolio Model

The model does not currently account for:

- transaction costs;
- taxes;
- liquidity;
- market impact;
- fundamental analysis;
- macroeconomic scenarios;
- dynamic rebalancing;
- investor-specific risk preferences.

10. Potential Improvements

Flexible Historical Periods

Allow users to select the analysis window, for example:

- 5 years
- 1 year
- 6 months
- 3 months

This would allow portfolio characteristics to be analyzed across different historical market environments.

User-Selected Assets

Allow users to select the securities included in the investment universe directly from the Dashboard. This would make the model more flexible and allow different investment universes to be tested without modifying the underlying model.

Multi-Asset Allocation

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Expand the investment universe to include:

- equities;
- ETFs;
- bonds;
- commodities;
- other asset classes.

This would transform the current equity-only model into a broader asset allocation framework.

Risk Aversion

Introduce a risk-aversion parameter and explicitly model the allocation between risky assets and the risk-free asset.

This would make the optimization more adaptable to different investor profiles.

11. Project Development

This project is **continuously evolving** as new financial models and analytical tools are developed.

The broader objective is to build a **set of automated tools covering different stages of the investment process**:



The current Portfolio Optimization & Asset Allocation Dashboard represents the asset allocation and portfolio construction stage of this framework.

Future developments may include:

- **Company Valuation & Security Selection** through automated Excel-based fundamental analysis and valuation tools with DCF model.
- **Investment Tracking & Monitoring** through portfolio performance and allocation monitoring.
- **Python Integration** for advanced statistical analysis, backtesting, risk modelling and quantitative analysis.

→ Therefore, note that additional models, tools and supporting files may be added to this project, with the objective of progressively building a broader **investment analysis and portfolio management toolkit**.